Updates for Supun: ML Workstream

Oct 7, 2026 · @Minidu

Your workstream is in good shape. Six things need changing now that you have Research Approach v3. The most urgent is the forecast JSON, because the data model is built from it on 9 Oct. Please push your code to the `Supun-Tharaka` branch by 8 Oct. The repo is public now, so access is no longer a problem.

Your corrections were right and have gone into the Research Approach (version 3.1): Mosavi, the Flood Hub API point value, the confirmed gauge level names, the HIGH / MEDIUM / LOW double meaning, and your scenario.

Read these first

Document	Where	What you need from it
Research Approach v3.1	Shared doc link from Minindu; repo `docs/research/research-approach-v3.md`	Task 2 (prediction types), Task 2b (comparison), Task 3 (the three profiles and the worked example), Task 5 (invariants)
XAI guideline evidence catalogue	Repo `docs/research/xai-guideline-evidence-catalogue.pdf`	The full wording of G11, G21, G25, G26, G38, G50, G73, G89, G119, G120 for re-rating
Prototype build plan	Repo `docs/prototype/build-plan.md`	File names, shared components, the "done when" checklist
Synthetic data README	Repo `ews-mobile/data/synthetic/README.md`	Rules for the forecast file
Screen list	Repo `docs/screen-specs/README.md`	Your screens S07–S11 and their guidelines

You do not need the task-1 triage files or the walkthrough files yet.

Answers to your open points

Your question	Answer
Per-value `synthetic` wrapper, or once per object?	Once per forecast object: `"synthetic": true`. The repo README already says so. Keep your generator check that fails if a forecast lacks it.
Thresholds: synthetic, or real with a source?	Keep 6.5 / 7.5 / 9.5 m flagged synthetic. The paper says the levels are "modelled on the Irrigation Department bulletin of 8 Aug 2026".
Both days are "Possible", so what does "what changed" show?	The class change (NO_FLOODING → ABOVE_NORMAL) and the picture change (1 of 10 → 3 of 10 houses), plus the direction ("went up").
Who supplies real landmark levels?	Nobody for this paper. Keep "the old ferry steps" and label it a placeholder.
The 5% cut between Unlikely and Possible	Keep it, and label it our own choice in a code comment and in the paper. CAP only says p ~ 0.
Action lines for levels other than "Possible"	Keep them, labelled placeholder, until a source is found (see section 6).

1. Scenario

Your scenario is now the team's scenario. Keep your numbers; nothing to change in the data. The Research Approach v3.1 now reads:

> Hanwella gauge, Kelani Ganga. Today ABOVE_NORMAL, with a 0.29 chance (about 30%) of reaching the Minor Flood Level on Thursday. Yesterday NO_FLOODING at 0.10. Both are "Possible" under CAP.

One line to add in your paper section: treating the Minor Flood Level as Flood Hub's danger threshold (SEVERE) is our own inference. You already say this for the three thresholds; say it once for the scenario too.

The paper's running example can keep "about 30%" and "3 in 10". Screens show 29% when the user asks for the exact number.

2. Forecast JSON (agree with Minindu by 9 Oct)

The data model types are written on 9 Oct from your JSON, so these changes come first.

1. **File and location.** Rename `forecast.synthetic.json` to `ews-mobile/data/synthetic/forecasts.json`, as the synthetic README says.
2. **Synthetic flag.** Drop the `{ "value": …, "synthetic": true }` wrapper. Put `"synthetic": true` once on each forecast object. Update `make_forecast_json.py` so its check looks for the object flag.
3. **Two full forecasts.** Replace the nested `previousForecast` with a list of two complete forecasts for the same gauge: yesterday (NO_FLOODING, 0.10) and today (ABOVE_NORMAL, 0.29). The app compares them itself. This is the "stored previous forecasts and a comparison" capability in the framework, so it must be real, not precomputed.
4. **Add `hazard`.** `"hazard": "flood"`. It is one of the five invariants, and today it is only implied by the gauge.
5. **Add `action`.** Move the action steps out of `explain.ts` into the JSON, as a list of steps, for example `["Get your go-bag ready today."]`. Action is an invariant too, so the invariants check needs it in the data.
6. **Keep everything else**: gauge, issue and next-update times, severity class, probability, CAP word, window, peak level low/median/high, 7-day levels, thresholds, landmark, supporting observations and `whatWouldChangeIt`.

One shared type joins your code and Minindu's: `explain.ts` returns an `AdaptedExplanation` with one text slot per invariant (hazard, place, time window, likelihood, action) plus optional extras. Minindu's `checkInvariants()` checks it. Your 28 tests then call that function, so the rule "the prediction never changes" is tested in one place.

3. Comparison table (re-rate by 11 Oct)

The table is the paper's motivating study, so a reviewer will check how the ratings were made. Three changes, then a blind comparison with Minindu.

1. **Add a "Hazard named (G11, G120)" row.** The Research Approach says to score each output against the five invariants, and hazard is the one invariant with no row. Raw SHAP, LIME and DiCE never say "flood"; they say "class 1" or "f(x)". That gives 8 needs × 3 methods = 24 cells.
2. **Re-rate every cell against the full guideline text**, in the evidence catalogue. You rated from the brief's short labels, and the full text can change a rating. For example, G21 is "Provide summary of long documents" and G73 is "Persistent visual context", which ask for more than "bottom line first" and "place and time". Keep Yes / Partly / No plus a one-line reason.
3. **Say which profile each "adapted" cell comes from.** Your last column mixes the three profiles. "It is not certain" is "Just tell me what to do"; the landmark and the reasons are "Show me why"; "Yesterday the chance was lower" is "How sure, and what changed?". Add the profile name in brackets in each cell. A reader should not think one screen shows all of it.

The blind second rating. Minindu rates the same 24 cells on 12 Oct from your raw outputs only, without seeing your ratings. Do not send him your new ratings before then. Then the two of you compare, count the cells where you agree, and settle the rest together. The paper reports it as "two authors rated independently and agreed on N of 24 cells before discussion", which is much stronger than one person's judgement.

Keep your "How to read the last column" notes. They are exactly right: the adapted column is a design proposal, untested with users, and its reasons say what the model relied on, never what causes a flood.

4. Screens and profiles (by 9 Oct; screens rebuilt after 22 Oct)

Your three placeholder profiles do not match the agreed ones. Replace `profiles.ts` with the three from Research Approach Task 3:

Agreed profile	Replaces your	What it bundles
"Just tell me what to do"	Same name	Main message first; step-by-step actions; words only; listen
"Show me why"	"Tell me why"	Reasons on; words + picture; landmark comparison
"How sure, and what changed?"	"Show me the details"	Words + picture + number; change alerts on

The profile type itself lives in Minindu's `model/profile.ts` from 9 Oct, so import it rather than keeping your own.

Leave the screen layouts as they are until Imansha's wireframes arrive on 22 Oct, as your doc already says. Two shared components are yours, from the build plan: `LikelihoodView` (CAP word, 10-house icon array, exact figure on tap, screen-reader label "3 of 10 houses filled") and `ForecastHeader` (place, time window, "Synthetic data" label). Tell Minindu your target date for both; his S03 and S04 screens use `LikelihoodView`.

5. Paper draft (by 20 Oct)

The draft is accurate but too long for its space. In the 8-page plan the motivating study gets about 1 page, and the prediction-type choice is one part of a 1.25-page section. Your two sections are about 900 words plus two tables and a figure.

1. **Cut to about 600 words in total**: about 200 for "What the model predicts" and 400 for "What standard XAI gives a resident".
2. **Keep one table and the figure.** Keep the comparison table and the raw-output figure. Drop the systems table and say its content in two sentences.
3. **Source or soften one claim.** "Little work addresses how to communicate it" needs a citation (Spiegelhalter et al., 2011, *Science* is in our evidence catalogue) or the label "our observation".
4. **Soften the counterfactual sentence.** You showed the unconstrained model gives 78.5% at 0.2 mm of rain. You did not show DiCE finding that point. Either run DiCE on the unconstrained model and report what it returns, or write "a counterfactual search could surface such an artefact".
5. **Add the hazard row** and the agreement count to the table once section 3 is done.

6. Checks before anything is cited

1. **LIME seeds.** Rerun LIME on the scenario row with 5 different random seeds. Report the range of the surrogate's prediction and of the top-ranked input. If the 0.58 holds across seeds, the paper can say so; if it moves a lot, report the range instead.
2. **References from memory.** Open the DOI or arXiv page for SHAP, LIME, Wachter, DiCE, Krishna (2022) and Caravan, and correct venue, pages and DOI. Wachter is *Harvard Journal of Law & Technology* 31(2); the PDF is on the journal site.
3. **Choubin et al. (2025).** Find the volume and article number, or drop it. The paper needs only one example of SHAP on a flood model.
4. **Action wording for Unlikely, Likely and Observed.** Find a source: Sri Lankan DMC public guidance, or the actions in our 128-guideline catalogue. Until then, label them placeholder in the JSON.
5. **MyShake alert pathway.** Minindu will send Competitive Analysis 03. Check your MyShake row against it.
6. **Gauge thresholds.** You found two sets of values for Hanwella. The paper uses the 8 Aug 2026 bulletin; cite that date and do not print the other set.

Deadlines

- [ ] 8 Oct: push the zip contents to the `Supun-Tharaka` branch
- [ ] 9 Oct: JSON changes (section 2) and `profiles.ts` (section 4); agree the `AdaptedExplanation` type with Minindu
- [ ] 11 Oct: re-rated comparison table with the hazard row (section 3); keep it to yourself until the 12 Oct comparison
- [ ] 12 Oct: LIME seeds rerun (section 6, item 1); compare ratings with Minindu
- [ ] 14 Oct: one-slide summary of the XAI comparison for the supervisor meeting on 15 Oct
- [ ] 20 Oct: trimmed paper sections (section 5) and the reference checks (section 6)
- [ ] After 22 Oct: rebuild S07–S10 from Imansha's wireframes